A pre-registered evaluation of behavioral weight edits in language models: metric floors, quantization anchors, and folding under post-normalization

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Abstract

Behavioral edits of language models are increasingly folded into the weights. We present a pre-registered evaluation protocol for such edits: target metrics are checked against corrupted models, damage is normalized by the harm of 4-bit (NF4) and 8-bit weight quantization, judges are validated against reference labels, and safety sets are sized to resolve the damage threshold. We apply it to a directional-ablation method for over-refusal on three models (Gemma-2-2B, Qwen2.5-1.5B, OLMo-2-1B) and to a steering-to-weights method for truthfulness on Gemma-2-2B. Without a degeneration check, a corrupted Gemma-2-2B would count as helpful on a third of labelled over-refusal prompts. No configuration of the over-refusal method is acceptable on Gemma-2-2B or OLMo-2-1B. On Qwen2.5-1.5B its gain exceeds that of KL-matched random directions, but NF4 quantization alone gives a gain of similar size (65–113% of the edit's gain over six extracted directions), and under a stricter, post-hoc safety threshold every extraction fails while a system prompt passes. On Gemma-2-2B, 200 safety prompts make the verdict on the mildest configuration close to a coin flip. For truthfulness, no gain is distinguishable from zero on the development split; on an already inspected test split, the published configuration gains 4.8 points [0.6, 9.1]. For models that normalize sublayer outputs before the residual addition, we derive the exact relation between folded and residual-stream ablation, show that the γ-compensated fold breaks when normalization gains are near zero, and give a pseudo-inverse fold that matches exact ablation within two points; the method's failures persist under exact ablation.

Keywords: model editing, activation steering, refusal, evaluation, RMSNorm, LLM safety

1. Introduction

Steering a language model's behavior along a direction in its hidden states is cheap and often effective (Turner et al., 2023; Rimsky et al., 2024; Zou et al., 2023a). For deployment, such interventions are increasingly folded into the weights, so that the edited model runs without hooks and can be shared, quantized and served like any other checkpoint: refusal can be removed by orthogonalizing the weights against a direction

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(Arditi et al., 2024), false refusal can be reduced by folding the ablation of a false-refusal direction (Wang et al., 2025), and steering vectors can be converted into rank-one edits of attention heads and neurons (Sun et al., 2026).

Whether such an edit is worth deploying depends on how it is evaluated, and several parts of a typical evaluation can make an edit look better than it is. A target metric can reward a broken model: a model that never refuses scores perfectly on a refusal-rate metric. Damage to other capabilities can be judged against a tolerance chosen by hand. Safety can be measured on too few prompts to resolve that tolerance. Judges may not have been checked against reference labels on the models being edited. And an improvement may not be compared with what a generic, behavior-agnostic perturbation of the same size would do.

We present an evaluation protocol for behavioral weight edits in which every decision was pre-registered on development data before the measurement it governs, and test data were measured once (Fig. 1). Its components are: a metric-floor check with deliberately corrupted models; a damage index that normalizes the harm on each capability and safety measure by the harm of 4-bit (NF4) and 8-bit (Q8_0) weight quantization, a perturbation that practitioners already accept; judges selected by agreement with reference labels and re-checked on every model; and safety sets large enough to resolve the damage threshold. We apply it to two published methods on two behaviors: the over-refusal edit of Wang et al. (2025) on Gemma-2-2B, Qwen2.5-1.5B and OLMo-2-1B, and the truthfulness edit of Steer2Edit (Sun et al., 2026) on Gemma-2-2B.

The protocol leads to conclusions that a standard evaluation would not reach. The over-refusal method has no acceptable configuration on Gemma-2-2B or OLMo-2-1B. On Qwen2.5-1.5B its gain exceeds that of KL-matched random directions, yet NF4 quantization alone gives a gain of similar size, and whether the edit beats quantization depends on the sample of prompts from which its direction is extracted. Every extraction fails once safety thresholds are set by measurement resolution rather than by quantization (a post-hoc analysis), while a system prompt passes. On Gemma-2-2B, whether the mildest configuration passes the safety threshold on 200 prompts is close to a coin flip. For truthfulness, the scoring used in prior work can rate corrupted models above intact ones, and no gain is distinguishable from zero on the pre-registered development split.

Two of the three models normalize each sublayer's output before adding it to the residual stream. On such models, folding an ablation into the output matrices does not remove the direction from the residual stream. We state the exact per-sublayer relation between a γ-compensated fold and residual-stream ablation, show that this fold breaks on real checkpoints whose normalization gains contain zeros or near-zeros, and give a pseudo-inverse fold with an error bound that is attained at zero threshold. The pseudo-inverse fold matches the exact ablation within two points of the target metric, and the method's failures persist under exact ablation, so they are not an artifact of folding.

Our contributions are: (i) a pre-registered evaluation protocol for behavioral weight edits, with code and every data split fixed by content hashes; (ii) its findings on two cases and three models, including

specificity controls (quantization, KL-matched placebos and re-extracted directions) and the dependence of the safety verdict on sample size; (iii) for post-norm models, the exact relation between folded and residual-stream ablation and a pseudo-inverse fold. The γ-compensated update for $c = 1$ was released earlier in an open-source toolkit (jmagly, 2026b); we credit it and claim no priority for it.

2. Related work

From activation steering to weight edits. Activation steering changes behavior by adding a direction to hidden states at inference time (Turner et al., 2023; Rimsky et al., 2024; Zou et al., 2023a), for example by shifting activations along truthfulness directions in a few attention heads (Li et al., 2023). Because such interventions must be applied at every forward pass, several methods turn them into weight edits: Arditi et al. (2024) orthogonalize the weights against a refusal direction, Wang et al. (2025) fold the ablation of a false-refusal direction into the weights, and Steer2Edit (Sun et al., 2026) converts steering vectors into rank-1 edits of individual attention heads and MLP neurons. Knowledge editing (Meng et al., 2022, 2023; Fang et al., 2025) also edits weights, but targets facts rather than behaviors.

Refusal directions and over-refusal. Refusal is mediated by a single direction in the residual stream of many chat models (Arditi et al., 2024). Wang et al. (2025) extract a false-refusal direction, orthogonalize it against the true-refusal direction, and ablate it, reporting lower false refusal with preserved safety and general capabilities. SCANS (Cao et al., 2025) steers activations for the same goal, and ProSafePrune (Chen et al., 2026) prunes weights; no code was available for the latter (Section 7). We use Wang et al. (2025) as the main baseline because it is a weight edit with released code.

Post-block normalization. The equivalence between ablating a direction and orthogonalizing the residual writers assumes that the writers' outputs enter the residual stream unnormalized. Practitioners have noticed that it fails on Gemma-2: a public "abliterated" Gemma-2-9B states that orthogonalization did not reproduce the runtime intervention because of the RMSNorm before the residual add (Gusev, 2024). Two weeks before our implementation, the OBLITERATUS toolkit added a fold of the post-sublayer norm scale for Gemma-2/3 and OLMo-2 (jmagly, 2026b); its update, $W \leftarrow W - (\Gamma^{-1}r)(r^\top \Gamma W)$, is the $c = 1$ case of the fold we analyze in Section 3, which we arrived at independently and learned of afterwards. That work reports, on Gemma-2-2B, the projection left on the removed direction and a behavioral reproduction of Tagliabue et al. (2026), whose reference implementation also removes the direction; it does not state the exact relation to residual-stream ablation, does not examine the hooks used in published evaluations, and does not measure capabilities or safety.

Evaluating steering and editing. Steerability varies widely across inputs, and for several concepts steering vectors are brittle to reasonable changes in the prompt (Tan et al., 2024); evaluations have relied on demonstrations rather than standardized metrics (Pres et al., 2024; Im and Li, 2025); and on a large benchmark, prompting outperforms representation-based steering (Wu et al., 2025), which is why we report a system-prompt baseline. Side effects of steering are common and structured (Ong et al., 2026; Malla et al., 2026; Bhandari et al., 2026; Siu et al., 2026), and steering, even along benign or random directions, can erode safety (Korznikov et al., 2025; Xiong et al., 2026). Knowledge edits degrade general abilities (Gu et al., 2024), gradually or catastrophically at scale (Gupta et al., 2024), and sometimes after a single edit (Yang et al., 2024). Our protocol borrows from the statistics of NLP evaluation: power analysis (Card et al., 2020) and pre-registration (van Miltenburg et al., 2021).

Quantization and safety. We use weight quantization as a reference perturbation. Its effect on safety is not neutral in general: on LLaMA2-13B-Chat, 4-bit GPTQ and AWQ quantization keeps each of eight trustworthiness dimensions, including toxicity, within five points of the original model, while 3-bit quantization degrades them (Hong et al., 2024); and a model can be constructed to behave benignly in full precision but maliciously once quantized with LLM.int8(), NF4 or FP4, for example by refusing benign requests (Egashira et al., 2024). On the smallest of our models we observe NF4 quantization lowering safety (Section 6.3); the anchor is therefore a useful reference for capabilities, but a lenient one for safety on such models.

3. Folding a directional ablation into the weights

Directional ablation. Let $r \in \mathbb{R}^d$ be a unit direction in the residual stream. Ablating r with coefficient $c \in [0, 1]$ maps every vector written to the residual stream to $(I - c\,rr^\top)y$ (Arditi et al., 2024; Wang et al., 2025). In a pre-norm block, the attention and MLP output matrices write to the stream directly, so the ablation can be folded into them, $W' = (I - c\,rr^\top)W$, and into the token embedding (Wang et al., 2025, Appendix D).

Post-block normalization. Gemma-2 (Gemma Team, 2024) and OLMo-2 (Team OLMo, 2025) normalize each sublayer's output before the residual add. With h the sublayer's input to its output matrix W,

$$x = Wh, \qquad y = \Gamma \frac{x}{\rho(x)}, \qquad \rho(x) = \left(\tfrac{1}{d}\textstyle\sum_i x_i^2 + \epsilon\right)^{1/2}, \qquad \Gamma = \operatorname{diag}(\gamma_{\text{eff}}), \tag{1}$$

where $\epsilon > 0$, $\gamma_{\text{eff}} = 1 + w$ in Gemma-2 and $\gamma_{\text{eff}} = w$ in OLMo-2, and y is what the sublayer adds to the residual stream. After the naive fold with $c = 1$, $x \perp r$, but $r^\top y = (\Gamma r)^\top x/\rho(x)$, which is not zero unless Γr is parallel to r. The folded model therefore keeps writing along r.

Proposition 1 (Gamma-compensated fold)**.** *In the setting of Eq.* (1)*, assume every entry of* γ_{eff} *is non-zero, let* $c \in [0, 1]$*, and let*

$$W' = \big(I - c\,\Gamma^{-1} r\, r^\top \Gamma\big)W. \tag{2}$$

For every input h, with $x = Wh$, $x' = W'h$, $y = \Gamma x/\rho(x)$ and $y' = \Gamma x'/\rho(x')$:

(i) $\Gamma x' = (I - c\,rr^\top)\,\Gamma x$;
(ii) $y' = \dfrac{\rho(x)}{\rho(x')}\,(I - c\,rr^\top)\,y$; *in particular* $r^\top y' = (1-c)\,\dfrac{\rho(x)}{\rho(x')}\,r^\top y$, *which is zero for* $c = 1$;
(iii) W' *is the only matrix* V *with* $\Gamma V h = (I - c\,rr^\top)\Gamma W h$ *for every* h.

Proof. (i) $\Gamma W' = \Gamma W - c\,r\,r^\top \Gamma W = (I - c\,rr^\top)\Gamma W$; apply both sides to h. (ii) Divide (i) by $\rho(x') > 0$ and use $\Gamma x = \rho(x)\,y$; the second statement follows from $r^\top(I - c\,rr^\top) = (1-c)\,r^\top$, since $\|r\| = 1$. (iii) If $\Gamma V h = (I - c\,rr^\top)\Gamma W h$ for every h, then $\Gamma V = (I - c\,rr^\top)\Gamma W = \Gamma W'$ as matrices, and $V = W'$ because Γ is invertible. □

The statement is exact per sublayer and for a given input: the folded write is the residual-stream ablation of the original write, multiplied by the positive per-token scalar $\rho(x)/\rho(x')$. Because that scalar differs from one, the inputs of later blocks drift from those of the residual-stream ablation, so the equivalence is not end-to-end. Part (iii) says that, among all linear changes of W, this one alone removes the fraction c of the component of Γx along r while leaving every orthogonal component unchanged; other updates that zero the component along r for $c = 1$, such as projecting the columns of W off Γr, also change orthogonal components. For a pre-norm block ($\Gamma = I$ and no normalization after W) the update reduces to the naive fold. The $c = 1$ update is the one released in the OBLITERATUS toolkit on 16 October 2026 (jmagly, 2026b,a), which predates our implementation; we make no claim of priority for it.

Zero and near-zero gains on real checkpoints. The assumption that every gain is non-zero fails on the checkpoints we use. In Gemma-2-2B, six of the 52 post-sublayer norms (after attention in layers 7 to 12) have entries with $\gamma_{\text{eff}} = 0$ exactly, so Γ^{-1} does not exist and the update produces non-finite weights. In OLMo-2-1B every gain is non-zero, but the smallest is 1.05×10^{-6} (Fig. 5). There the update is defined but ill-conditioned: coordinate i of x' changes by $c\,|r_i|\,|r^\top \Gamma x|/|\gamma_i|$, so $\rho(x')$ can become much larger than $\rho(x)$, and the factor $\rho(x)/\rho(x')$ in (ii) then shrinks the whole write. The behavioral result is a collapse of the model (Section 6.4); the mean deviation of the write from the exact ablation is 17.5% (Fig. 5).

A pseudo-inverse removes both problems. For a relative threshold $\varepsilon \geq 0$, let $K = \{i : \gamma_i \neq 0,\ |\gamma_i| \geq \varepsilon \max_j |\gamma_j|\}$, let P_ε be the diagonal projection onto the coordinates in K, and let Γ_ε^+ be the diagonal matrix with entries $1/\gamma_i$ for $i \in K$ and 0 otherwise.

Proposition 2 (Pseudo-inverse fold)**.** *In the setting of Eq.* (1)*, let* $c \in [0,1]$, $\varepsilon \geq 0$, *and*

$$W' = \big(I - c\,\Gamma_\varepsilon^+ r\,r^\top \Gamma\big) W. \tag{3}$$

For every input h, with x, x', y, y' as in Proposition 1:

(i) $\Gamma x' = (I - c\,P_\varepsilon rr^\top)\,\Gamma x$, *hence* $y' = \dfrac{\rho(x)}{\rho(x')}\Big[(I - c\,rr^\top)\,y + c\,(I - P_\varepsilon)\,r\,(r^\top y)\Big]$;

(ii) before the per-token factor, the deviation from the residual-stream ablation is $c\,(I-P_\varepsilon)r\,(r^\top y)$*, of norm* $c\,\|(I-P_\varepsilon)r\|\,|r^\top y|$*;*

(iii) let $\varepsilon = 0$*,* $c > 0$ *and* $r^\top \Gamma W \neq 0$*. A matrix* V *with* $\Gamma V h$ *a positive multiple of* $(I - c\,rr^\top)\Gamma W h$ *for every* h *exists if and only if* $(I - P_0)r = 0$*, and* W' *is then such a matrix;*

(iv) let $\varepsilon = 0$*. For every matrix* V *and every* h*,* $\|\Gamma V h - (I - c\,rr^\top)\Gamma W h\| \geq c\,\|(I - P_0)r\|\,|r^\top \Gamma W h|$*. The fold* W' *attains this bound for every* h*, and among all matrices that attain it for every* h*,* W' *is the unique one that minimizes* $\|V - W\|_F$*.*

Proof. (i) $\Gamma\Gamma_\varepsilon^+ = P_\varepsilon$, so $\Gamma W' = \Gamma W - c\,P_\varepsilon r\,r^\top \Gamma W = (I - c\,rr^\top)\Gamma W + c\,(I - P_\varepsilon)r\,r^\top \Gamma W$. Apply to h, divide by $\rho(x') > 0$ and use $\Gamma x = \rho(x)\,y$. (ii) follows from (i). (iii) If $(I - P_0)r = 0$, then (i) gives $\Gamma x' = (I - c\,rr^\top)\Gamma x$. Conversely, let $i \notin K$ with $r_i \neq 0$, and choose h with $r^\top \Gamma W h \neq 0$. Since $\gamma_i = 0$, coordinate i of $\Gamma V h$ is zero for every V, while coordinate i of $(I - c\,rr^\top)\Gamma W h$ is $-c\,r_i\,r^\top \Gamma W h \neq 0$; no positive multiple matches it. (iv) On a coordinate $i \notin K$, $(\Gamma V h)_i = 0$ and the target is $-c\,r_i\,r^\top \Gamma W h$; summing squares over these coordinates gives the bound. By (i), $\Gamma W' h$ equals the target on K and is zero outside K, so W' attains it. A matrix V attains the bound for every h only if $\gamma_i (Vh)_i$ equals the target on every $i \in K$ for every h, which fixes each row $V_{i\cdot}$, $i \in K$, to $W'_{i\cdot}$. The rows outside K are free, and $\|V - W\|_F$ is minimized only by leaving them equal to those of W, which is what W' does because $(\Gamma_0^+ r)_i = 0$ for $i \notin K$. □

The optimality in (iv) concerns the write before the per-token normalization factor. The coordinates with $\gamma_i = 0$ do not reach the residual stream but still enter $\rho(x')$; the pseudo-inverse fold leaves them unchanged, and we make no claim about how close $\rho(x')$ is to $\rho(x)$. A threshold $\varepsilon > 0$ also drops gains that are small but non-zero: this trades a leak bounded by (ii) for avoiding the growth of $\rho(x')$ that silences the γ-compensated fold. We choose ε per model by a pre-registered rule that uses only forward passes of the unedited model (Section 5). On a pre-norm block the pseudo-inverse fold reduces to the naive fold.

Tied embeddings. Every fold also folds the token embedding, naively, since no normalization follows it. Gemma-2-2B and Qwen2.5-1.5B tie the embedding to the output head (OLMo-2-1B does not), so on these models every fold also changes the output logits (before Gemma-2's final soft-capping), by $-c\,(Er)(r^\top \tilde{h})$ for embedding matrix E and final hidden state $\tilde{h}$, whereas hooks leave the head unchanged. Differences between a fold and hooks on Gemma-2 and Qwen2.5 therefore combine the effect of the normalization with that of the tied head. The pseudo-inverse and naive folds treat the embedding identically, so their difference isolates the normalization; on Qwen2.5 we also measured a fold of the output matrices alone, with the embedding ablated by a hook (Section 6.4).

Where the hooks of the released code ablate. The released code of Wang et al. (2025) evaluates the ablation with hooks (`pipeline/utils/hook_utils.py`, lines 80–90, commit `916dcac`): the direction is removed from the input of every decoder block and from the outputs of every attention and MLP module. The repository

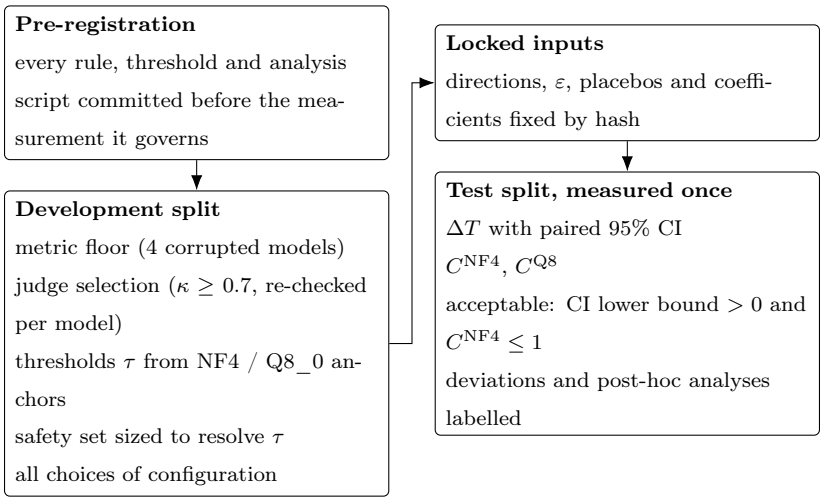


Figure 1: The evaluation protocol. Every decision is taken on development data under rules written in advance; test data are measured once per configuration with inputs fixed by content hashes.

ships model wrappers for Llama-2 and Llama-3, which are pre-norm; there, these module outputs are exactly the residual-stream writes, and the hooks implement the intended ablation. On a post-norm model the same placement removes r before the normalization. The block-input hooks still clean the residual stream at the start of every block, so less leaks than with the naive fold: the component along r reintroduced by the attention sublayer's norm reaches the MLP of the same block, and the one written by the last block reaches the output. We call this placement, ported to Hugging Face Gemma-2 and OLMo-2, the *released hooks*, and compare it with hooks on the residual-stream writes (after the norm), which implement the ablation exactly. This concerns the use of the method on post-norm models, which Wang et al. (2025) did not evaluate. All variants are compared in Section 6.4.

4. Evaluation protocol

An edit is judged on two axes: the gain on the targeted behavior, ΔT, and the damage it does elsewhere. The protocol below fixes how both are measured before any edit is scored. Its parts address four ways in which the evaluation of behavioral edits can mislead: a metric that rewards broken models, a damage threshold that is set arbitrarily or that rewards small samples, judges that have not been checked, and safety measured on too few prompts.

4.1. Pre-registration and the use of test data

Every design decision, threshold and analysis was written into a time-stamped pre-registration file under version control before the measurement it governs, and every analysis script was committed before it was first run. Deviations are listed in Appendix A. Every data set is split into a development part, used for all

decisions, and a test part, measured once per configuration after all decisions are fixed. Each split is fixed by row indices and by the SHA-256 hash of its rows in canonical JSON; jobs refuse to run when a hash does not match. One exception is disclosed: the TruthfulQA test split of the truthfulness case was seen during the reproduction of Steer2Edit and the metric-floor diagnostics (Section 5), so the truthfulness test numbers are descriptive.

4.2. Vitals and the normalized damage index

Damage is measured on a fixed panel of capability measures, which we call vitals: accuracy on MMLU (Hendrycks et al., 2021) and ARC-Challenge (Clark et al., 2018), the negative log-likelihood (NLL) of WikiText-2 (Merity et al., 2017), the NLL of GSM8K reference solutions (Cobbe et al., 2021), the log-probability of the end-of-turn token after those solutions, and the rate of greedy GSM8K solutions that finish within 512 tokens. GSM8K accuracy on 46 generated solutions is reported but not used, since one item moves it by 2.2 points. In the over-refusal case three safety vitals are added (Section 4.5). For vital j, the harm h_j is the mean paired difference between the unedited and the edited model, signed so that a positive value means damage.

Vitals are in different units, and a fixed tolerance such as "at most one point" is arbitrary. We therefore normalize each harm by a threshold tied to a perturbation practitioners already accept: weight quantization. At level L, the anchor is the model with every linear layer of the decoder blocks quantized and dequantized, run in the same fp32 pipeline as every edit. The primary level is 4-bit NF4 with the QLoRA codebook, blocks of 64 and absmax scales (Dettmers et al., 2023); our implementation matches bitsandbytes (Dettmers and bitsandbytes contributors, 2026) bit for bit. The strict level is the 8-bit Q8_0 format of llama.cpp (Gerganov and llama.cpp contributors, n.d.), with blocks of 32 and fp16 scales. Only the weights are perturbed: LLM.int8 (Dettmers et al., 2022) is not used because it also changes how outlier activations are computed. The threshold of vital j is

$$\tau_j^L = \max\big(h_j^L,\, r_j^L\big), \tag{4}$$

where h_j^L is the harm of the anchor and r_j^L the measurement resolution, twice the bootstrap standard error of the paired difference between the unedited model and the anchor. For vitals measured as rates, r_j^L is at least two items ($200/n_j$ points), so that an anchor that changes almost no answers does not produce a threshold near zero. The damage index is

$$C^L = \max_j\, h_j/\tau_j^L, \tag{5}$$

and a configuration is acceptable at level L when $C^L \le 1$, that is, when no vital is damaged more than by the anchor or than the measurement can resolve. Thresholds are fixed once per model, before any edit is measured, and always on the full development part of each vital. We considered and rejected a threshold derived from the standard error of the edit itself: it loosens as the sample shrinks, and so rewards measuring

on fewer items. Taking the maximum over vitals makes the index conservative; we report a near-harmless perturbation as a negative control (Section 6.3). The end-of-turn vital is kept in C; C without it is reported as a post-hoc sensitivity analysis.

4.3. Target metric and the metric floor

A metric that only asks whether a model refused rewards a broken model, which never refuses. The over-refusal target is therefore the rate of *helpful compliance* T: a reply counts when it answers the request, and not when it refuses (fully or partly) or is off-topic or degenerate. Every target metric is checked against a floor: four deliberately corrupted models (Gaussian noise added to every decoder linear weight at 0.3 and 1.0 times the weight's standard deviation, zeroed attention output projections, and re-initialized decoder linear weights) must score $T \leq 5\%$. The judge used for T (below) has no label for degenerate text, so a reply counts as helpful only if it also passes a degeneration gate adapted from Ong et al. (2026): (a) at least 30% of its token trigrams are distinct (replies under 10 tokens are exempt); and (b) its perplexity under the unedited model, conditioned on the prompt, is at most 1.2 times the 95th percentile of the perplexities of the unedited model's own replies. A variant of (b) conditioned on an empty user turn agreed better with the reference labels but failed the floor check (one corrupted model reached $T = 7.5\%$), so the pre-registered rule kept the prompt-conditioned gate.

4.4. Judges validated against reference labels

Judges were chosen by a rule fixed in advance: among the candidates, the one with the highest Cohen's κ (Cohen, 1960) on the binary quantity that the metric counts, provided $\kappa \geq 0.7$, chosen before any edited configuration was scored. The candidates were an LLM judge with a three-label rubric (Qwen3-4B-Instruct-2507), Qwen3Guard-Gen-4B (Zhao et al., 2025), and the refusal substring list of Wang et al. (2025).

Reference labels for 100 replies to benign prompts and 80 replies to harmful prompts were produced by two LLM annotators from model families different from the candidate judges (Gemini and ChatGPT, accessed through their public chat interfaces), each in a fresh conversation, with neutral instructions that did not mention the study, on replies whose source configuration was hidden. The replies were sampled with a fixed seed from the unedited model, edited configurations and corrupted models. The annotators agreed on 97/100 benign and 79/80 harmful replies (on the first 130: binary $\kappa = 0.94$ and 0.93); the author adjudicated the disagreements. As a pre-registered check, the author labeled 30 randomly drawn benign replies blind to both annotators and to the source; 29 of 30 matched the reference (threshold: 27).

The selected judges, all on Gemma-2-2B replies, are: for T, Qwen3Guard's refusal field combined with the degeneration gate ($\kappa = 0.76$; Qwen3Guard alone 0.86); for refusal of harmful prompts, the substring list ($\kappa = 0.84$, sensitivity 0.91, specificity 0.96); for harmful content, Qwen3Guard's *Unsafe* or *Controversial* label on a non-repetitive reply ($\kappa = 0.905$, sensitivity 0.95, specificity 0.97). The last judge replaced *Unsafe*

alone ($\kappa = 0.61$) after Qwen3Guard was found to label all six degenerate replies among the first 30 references as unsafe. Because judge agreement can differ across the models being judged, the judges were re-checked on fresh reference labels for each additional model. On OLMo-2-1B all three judges passed ($\kappa = 0.733$, 0.875 and 0.828). On Qwen2.5-1.5B the judge for T (0.933) and the harmful-content judge (1.000) passed, and the refusal judge did not (0.615). Every quantity on Qwen2.5-1.5B that uses the refusal judge, including the harmful-refusal vital and therefore C, is reported as descriptive (Appendix A, S9).

4.5. Safety on an adequate sample

An edit that reduces over-refusal must not increase compliance with harmful requests. Three safety vitals enter C: the refusal rate and the harmful-content rate on harmful prompts (judges above), and a continuous refusal score, $\log p(\text{“I”}) - \log(1 - p(\text{“I”}))$ at the first reply position, the quantity used to select refusal directions (Arditi et al., 2024; Wang et al., 2025). The development set has 1,031 harmful prompts (all of OR-Bench-Toxic (Cui et al., 2025) and the AdvBench (Zou et al., 2023b) prompts that are neither in the direction-extraction pool nor in HarmBench, after removing near-duplicates at word-set Jaccard ≥ 0.7). The test set has 504 prompts from HarmBench (Mazeika et al., 2024), JailbreakBench (Chao et al., 2024) and StrongREJECT (Souly et al., 2024), deduplicated against each other and against all development and extraction prompts. The size matters: with a base harmful-content rate of a few percent, one reply moves the rate by half a point on 200 prompts, which is the size of the threshold itself (Section 6.2).

4.6. Statistics

All comparisons are paired over items, with 10,000 bootstrap resamples and 95% percentile intervals. When an interval contains zero we say the difference is not detected. Decoding is greedy without repetition penalty: 256 new tokens for target replies, 512 for GSM8K. Each configuration is run once; the variance across samples of the prompts from which a direction is extracted was measured for one configuration (Section 6.3). Every table, figure and number in Sections 6 and 7 is produced by one script with fixed seeds from the saved replies and labels; its point estimates equal those of the registered analyses, its interval bounds differ from them by at most 0.4 points, and its values of C by at most 2% (the resolution r is a bootstrap standard error).

5. Experimental setup

Models. Gemma-2-2B-it (Gemma Team, 2024) and OLMo-2-0425-1B-Instruct (Team OLMo, 2025) normalize the attention and MLP outputs before the residual add ($\gamma_{\text{eff}} = 1 + w$ and $\gamma_{\text{eff}} = w$; OLMo-2 has no input norm and normalizes queries and keys); Qwen2.5-1.5B-Instruct (Qwen Team, 2024) is a pre-norm model and serves as the control for the mechanism of Section 3. All models are pinned to a Hugging Face revision,

their weight files are hashed, and everything runs in fp32 on NVIDIA T4 GPUs (two per job). The judges run in fp16.

Over-refusal case. OR-Bench-Hard-1K (Cui et al., 2025) (1,319 prompts) is split in half at random, stratified by category; one half is the pool from which every edit is built, the other provides 400 development prompts for T. The test set for T is the 250 safe prompts of XSTest (Röttger et al., 2024), a different source. The main baseline is the method of Wang et al. (2025), reimplemented for Hugging Face decoders from their released code: a false-refusal direction and a true-refusal direction are extracted from mean activation differences at the post-instruction template positions (128 over-refusal, 128 harmful and 128 harmless prompts, filtered by refusal score as in their default settings); the false-refusal direction is orthogonalized against the true-refusal one with weight λ; the direction is selected on 32 held-out prompts (lowest over-refusal score with KL ≤ 0.2 on harmless prompts, last 20% of layers excluded). The selected direction r is ablated with coefficient c, in one of four ways: the naive fold of their Appendix D (embedding and every attention and MLP output matrix, $W' = (I - c\,rr^\top)W$), the γ-compensated fold of Proposition 1, the released hooks (ablation at every block input and every attention and MLP output, before the post-norm), and hooks on the residual-stream writes (after the post-norm), which is the edit the method intends. Units are named `wang:`λ`:`c for the naive fold. The development grid is $\lambda \in \{1, 0.8, 0.5\}$ and $c \in \{1, 0.5\}$. A fixed system prompt that asks the model to answer legitimate requests on sensitive topics is the non-editing baseline:

> You are a helpful assistant. Many requests that mention sensitive topics are legitimate, for example questions about safety, history, fiction, or how something works. Answer such requests helpfully. Decline only when a request clearly seeks to cause harm.

Gemma's chat template has no system role, so the text is prepended to the user turn.

Variants added on post-norm models. The pseudo-inverse fold of Proposition 2 was registered after the development results of the other variants and measured on both post-norm models at $\lambda = 1$ with $c \in \{1, 0.5\}$. Its threshold ε was chosen per model, before any of its outputs, as the value on the grid $\{0, 10^{-8}, \ldots, 10^{-1}\}$ that minimizes the mean relative distance between its post-norm write and the exactly ablated write, over 100 windows of 128 tokens of WikiText-2 training text, every post-norm sublayer and $c \in \{0.5, 1\}$, using the Wang direction at $\lambda = 1$. No behavioral metric or vital enters the choice.

Specificity controls. NF4 quantization is reported as an edit in its own right. Five placebo directions were drawn on Qwen2.5-1.5B from $\mathcal{N}(0, \Sigma)$, with Σ the sample covariance of the block inputs at the layer and token position of the Wang direction at $\lambda = 1$, normalized, and folded exactly like it. The coefficient of each was set by bisection so that its KL divergence from the unedited model on a reference text matches that of the configuration under test (`wang:0.8:0.5`, KL 0.0086) within 5%; the achieved gaps were between -2.6% and $+1.6\%$. The pre-registered rule calls the edit specific to over-refusal only if the paired intervals of both

ΔT(edit) − ΔT(NF4) and ΔT(edit) − ΔT(largest placebo) exclude zero (an intersection–union test, which needs no multiplicity correction). The placebo directions and coefficients chosen on development data were reused on test, checked by hash. After a first review of the results, we registered and ran a re-extraction of the direction: the three training pools (128 prompts each) were resampled with replacement five times, the selection procedure and its 32 + 32 held-out prompts were kept, and each resulting direction was folded with $\lambda = 0.8$ and $c = 0.5$ and measured on test.

Truthfulness case. Steer2Edit (Sun et al., 2026) was reimplemented from its paper and repository (which has no license, so no code was copied) and run on Gemma-2-2B-it with their TruthfulQA (Lin et al., 2022) split. Their training split (408 questions) was divided into 250 questions for extracting steering vectors, from Gemma-2-2B's own answers and the truthfulness labels released with their code, and 158 development questions. The target metric is the accuracy of choosing the correct of two answers by length-balanced pointwise mutual information (PMI) between answer and question, chosen after the metric-floor check (Section 6.5). Of the 25 configurations of their refined grid for this model, which edits attention heads only, the 23 that fitted the compute budget were measured on development data. The test split of TruthfulQA had been seen during the reproduction, so test numbers for this case are descriptive.

Test protocol. Each model is measured once on test, after all development decisions: the unedited model, the two anchors, the system prompt, and the method of Wang et al. (2025) at $\lambda = 1$ with $c = 1$ (their published setting) and $c = 0.5$, in all four variants on the post-norm models and in two (naive fold and released hooks) on Qwen2.5, plus every configuration that was acceptable on development data (`wang:0.8:0.5` on Qwen2.5). A later, pre-registered add-on measured the pseudo-inverse fold with the development ε and the five development placebos with their development coefficients. The directions are the development ones, checked by hash. Thresholds τ come from the anchors measured on the test parts; the perplexity limit of the degeneration gate is the development one. The primary endpoint is ΔT on XSTest with its interval and C^{NF4}.

6. Results

We report, in order, the over-refusal case on three models, the role of the size of the safety set, the specificity of the effect on Qwen2.5-1.5B, folding on post-norm models, the truthfulness case and the system-prompt baseline. Confirmatory results follow pre-registered rules; descriptive and post-hoc results are labelled as such.

6.1. Over-refusal on three models

Metric floor. With the degeneration gate, all four corrupted versions of each model scored $T = 0\%$ on the 40 labelled development prompts, below the 5% floor. The gate is necessary: judged by Qwen3Guard alone,

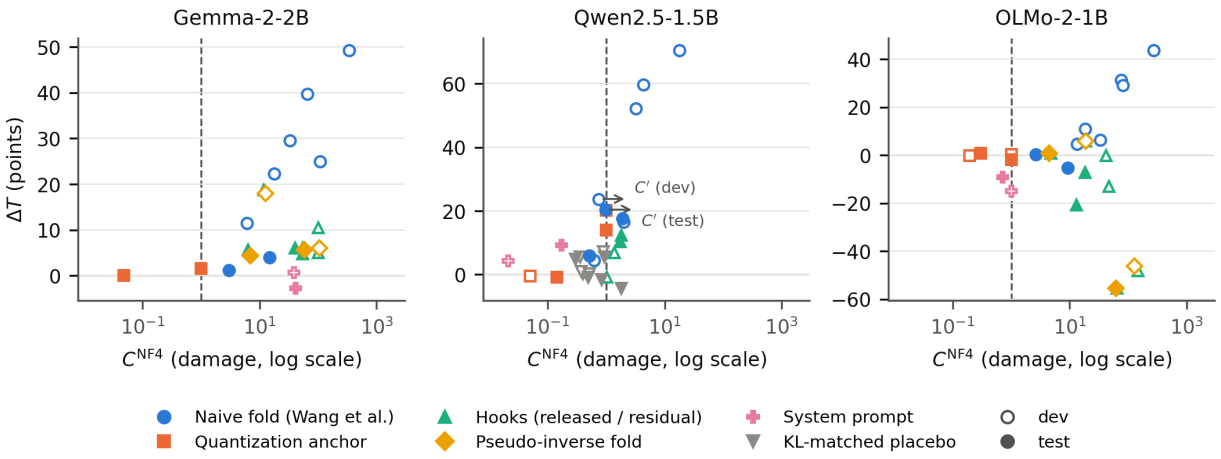


Figure 2: Gain on the target metric against the NF4-normalized damage index, for every measured unit (hollow: development; filled: test). Units left of the dashed line are within the NF4 threshold. Arrows on Qwen2.5-1.5B show C', the post-hoc index with safety thresholds at the measurement resolution, for `wang:0.8:0.5`. The γ-compensated fold is omitted (non-finite on Gemma-2-2B; collapsed on OLMo-2-1B, Section 6.4). C on Qwen2.5-1.5B is descriptive.

the noise-corrupted Gemma-2-2B (0.3σ) counted as helpful on 13 of the 40 prompts (32.5%).

Development. On Gemma-2-2B every configuration of the naive-fold grid raises T from 43.2%, with intervals above zero, and none is acceptable: C^{NF4} ranges from 6.1 to 337. The safety vitals decide: for every configuration the refusal score has the largest ratio h_j/τ_j, and the refusal and harmful-content rates on the 1,031 harmful prompts also exceed their thresholds. On OLMo-2-1B ($T_0 = 54.0\%$) no configuration of the method is acceptable either ($C^{\mathrm{NF4}} \geq 13$). On Qwen2.5-1.5B ($T_0 = 25.0\%$), `wang:0.8:0.5` raises T by 23.8 points [19.5, 28.2] with $C^{\mathrm{NF4}} = 0.74$, and `wang:1.0:0.5` by 4.5 [1.5, 7.5] with 0.63 (descriptive, Section 4.4). Table A.4 lists every unit.

Test. Table 1 and Fig. 2 give the primary endpoint. No configuration is acceptable on Gemma-2-2B or OLMo-2-1B, in any variant. On OLMo-2-1B the only unit with $C^{\mathrm{NF4}} \leq 1$ is the system prompt, which lowers T by 9.2 points. XSTest leaves little headroom on these two models: the unedited models are not helpful on only 16.4% (Gemma-2-2B) and 9.6% (OLMo-2-1B) of its prompts. The failures are on the damage axis, however, with C^{NF4} between 2.7 and 77.7 for every configuration of the method. On Qwen2.5-1.5B, `wang:0.8:0.5` replicates its development gain, $\Delta T = +20.4$ [15.2, 26.0] with $C^{\mathrm{NF4}} = 0.98$ (descriptive). NF4 quantization alone raises T by 14.0 points [8.8, 19.6].

6.2. The size of the safety set decides the verdict

On Gemma-2-2B, the method of Wang et al. (2025) raises helpful compliance on the 400 development prompts at every configuration of the grid; the mildest, $\lambda = 1$ and $c = 0.5$, by $\Delta T = +11.5$ points [8.2,

Table 1: Over-refusal, test endpoint. ΔT: change in the share of benign prompts answered helpfully (T_0: unedited model). C: damage index normalized by NF4 or Q8_0 quantization; C' (post hoc): safety thresholds at the measurement resolution only. Acceptable: CI lower bound > 0 and $C^{\mathrm{NF4}} \leq 1$. †Descriptive: the refusal judge failed validation on Qwen2.5-1.5B.

Edit	ΔT [95% CI]	C^{NF4}	C^{Q8}	C' (post hoc)	Acc.
Gemma-2-2B (T_0 = 83.6%)					
NF4 (anchor)	+1.6 [−0.8, +4.0]	1.00	23.4	1.00	–
Q8_0 (anchor)	+0.0 [−1.2, +1.2]	0.05	0.75	0.05	–
System prompt	−2.8 [−6.0, +0.4]	40.9	675	40.9	no
Naive fold (λ=1.0, c=1.0)	+4.0 [+0.4, +7.6]	14.9	226	14.9	no
Naive fold (λ=1.0, c=0.5)	+1.2 [−1.2, +3.6]	3.03	46.2	3.03	no
Released hooks (λ=1.0, c=1.0)	+6.0 [+2.4, +9.6]	54.6	832	54.6	no
Released hooks (λ=1.0, c=0.5)	+6.0 [+2.4, +9.6]	40.1	611	40.1	no
Residual hooks (λ=1.0, c=1.0)	+4.8 [+1.2, +8.4]	54.2	827	54.2	no
Residual hooks (λ=1.0, c=0.5)	+5.6 [+2.4, +8.8]	6.35	96.9	6.35	no
Pseudo-inverse fold (λ=1.0, c=1.0)	+5.6 [+1.6, +9.6]	57.7	879	57.7	no
Pseudo-inverse fold (λ=1.0, c=0.5)	+4.4 [+1.6, +7.2]	7.03	107	7.03	no
Qwen2.5-1.5B (T_0 = 54.4%)					
NF4 (anchor)	+14.0 [+8.8, +19.6]	1.00	269	13.9	–
Q8_0 (anchor)	−0.8 [−3.2, +1.2]	0.14	1.00	0.14	–
System prompt	+9.2 [+4.0, +14.4]	0.17	1.79	0.17	yes†
Naive fold (λ=1.0, c=1.0)	+17.6 [+12.4, +23.2]	1.90	424	21.9	no
Naive fold (λ=1.0, c=0.5)	+6.0 [+2.8, +9.2]	0.52	141	7.27	yes†
Naive fold (λ=0.8, c=0.5)	+20.4 [+15.2, +26.0]	0.98	36.7	3.31	yes†
Released hooks (λ=1.0, c=1.0)	+12.4 [+6.8, +18.0]	1.80	275	14.2	no
Released hooks (λ=1.0, c=0.5)	+10.4 [+5.2, +15.6]	1.75	111	5.71	no
Placebo 4	+5.6 [+1.6, +9.6]	0.94	206	10.7	yes†
OLMo-2-1B (T_0 = 90.4%)					
NF4 (anchor)	−2.0 [−4.8, +0.4]	1.00	37.8	2.61	–
Q8_0 (anchor)	+0.8 [−0.8, +2.4]	0.30	1.00	0.35	–
System prompt	−9.2 [−12.8, −5.6]	0.72	1.89	0.72	no
Naive fold (λ=1.0, c=1.0)	−5.2 [−8.8, −1.6]	9.26	136	14.5	no
Naive fold (λ=1.0, c=0.5)	+0.4 [−1.2, +2.0]	2.66	39.1	3.86	no
Released hooks (λ=1.0, c=1.0)	−20.8 [−26.4, −15.2]	13.0	217	23.4	no
Released hooks (λ=1.0, c=0.5)	−7.2 [−11.2, −3.2]	18.4	271	20.5	no
Residual hooks (λ=1.0, c=1.0)	−55.6 [−62.0, −49.2]	64.9	956	122	no
Residual hooks (λ=1.0, c=0.5)	+0.8 [−1.2, +2.8]	4.75	69.9	6.86	no
Pseudo-inverse fold (λ=1.0, c=1.0)	−55.6 [−62.0, −49.2]	61.5	905	108	no
Pseudo-inverse fold (λ=1.0, c=0.5)	+0.8 [−1.2, +2.8]	4.41	64.8	6.83	no
γ-comp. fold (λ=1.0, c=1.0)	−88.0 [−92.0, −83.6]	77.7	1143	115	no
γ-comp. fold (λ=1.0, c=0.5)	−78.8 [−84.4, −73.2]	38.0	340	87.4	no

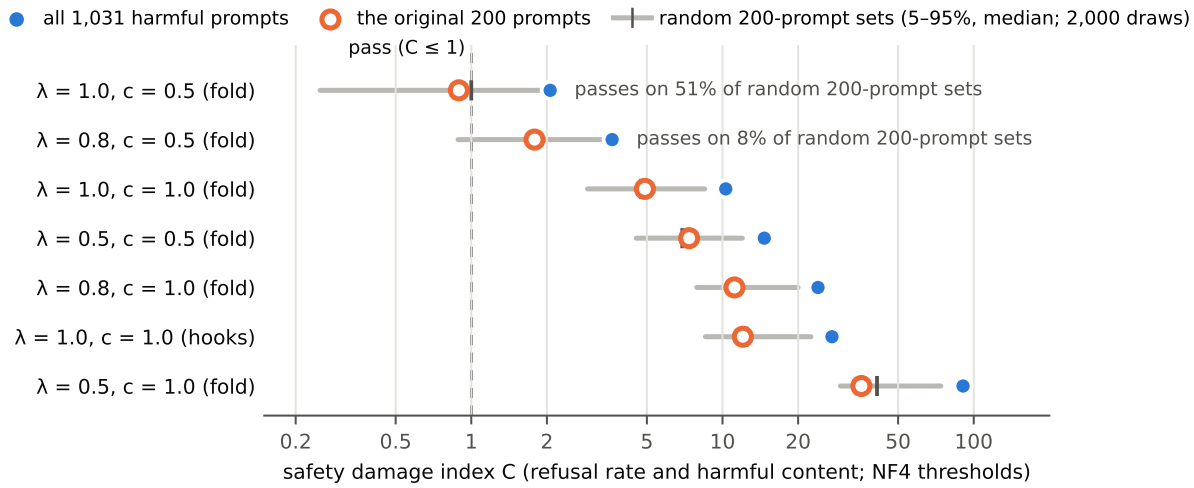


Figure 3: Safety damage index of the configurations of Wang et al. (2025) on Gemma-2-2B (development data), measured on all 1,031 harmful prompts, on the 200 prompts of an earlier stage, and on 2,000 random 200-prompt subsets (5–95% range and median). The index is the larger of the harm to the refusal rate and to the harmful-content rate, each divided by its NF4 threshold recomputed on the same prompts; a configuration passes when it is at most one. All points use the same replies and the same judges. Post-hoc analysis.

15.0]. Whether it is acceptable depends on how many harmful prompts are used to measure safety. With the registered set of 1,031 prompts, no configuration passes: at the mildest, the harmful-content rate rises by 1.65 points [0.78, 2.62] against a threshold of 0.80, so $C^{\mathrm{NF4}} = 2.06$ on this vital alone. With the 200 prompts used in an earlier stage of this study, the same configuration had passed.

Two things changed between those measurements: the number of prompts and the harmful-content judge. A post-hoc analysis separates them (Fig. 3): all prompt sets are scored with the final judges on the same replies, and the original 200 prompts are a subset of the 1,031. With the final judges, the mildest configuration still passes on the original 200 ($C_{\mathrm{safety}} = 0.89$) and fails on all 1,031 (2.06); over 2,000 random sets of 200 prompts it passes on 51%. The verdict follows the sample size, not the judge. The reason is the resolution floor of the threshold: with 200 prompts a harm smaller than about 2.2 points cannot be told apart from the anchor's noise, and with 1,031 prompts the floor drops to 0.8 points. On Gemma-2-2B, for its mildest configuration, the pass or fail on a 200-prompt set is close to a coin flip.

6.3. Specificity of the effect on Qwen2.5-1.5B

Because NF4 quantization alone raises T on Qwen2.5-1.5B, we asked whether the edit acts on over-refusal or adds a generic perturbation.

The registered comparison. By the pre-registered rule, on test the gain of `wang:0.8:0.5` exceeds both that of NF4 (+6.4 points [2.0, 10.8]) and that of the KL-matched placebo with the largest gain (+14.8 [10.0,

Table 2: Re-extraction of the over-refusal direction on Qwen2.5-1.5B (test; $\lambda = 0.8$, $c = 0.5$). Seed – is the development direction; seeds 1–5 resample the training prompts with replacement. C' is post hoc. The first row is the direction of Table 1, where its C' is 3.31; the last digit differs because the resolutions are bootstrap estimates.

Seed	Layer	cos	ΔT [95% CI]	NF4 share	C^{NF4}	C'
–	21	1.00	+20.4 [+15.2, +26.0]	0.69	0.98	3.30
1	21	0.99	+21.6 [+16.4, +27.2]	0.65	1.00	3.46
2	21	0.99	+16.4 [+11.6, +21.6]	0.85	0.81	2.73
3	21	0.99	+18.8 [+14.0, +24.0]	0.74	0.88	2.97
4	21	0.99	+13.2 [+8.8, +18.0]	1.06	0.71	2.41
5	21	0.98	+12.4 [+8.0, +17.2]	1.13	0.71	2.41

19.6]). On development data the difference from NF4 has the same sign but its interval includes zero (+3.5 [−0.8, 7.8]), so the rule's verdict there is "non-specific"; the difference from the largest placebo is +16.5 [12.8, 20.5].

The size of the effect (post hoc). NF4 quantization alone reaches 0.85 [0.69, 1.03] of the edit's gain on development data and 0.69 [0.48, 0.89] on test (Fig. 4). The largest placebo reaches about 0.3.

Re-extracted directions. The direction in these comparisons comes from one sample of extraction prompts. Resampling those prompts five times (registered before running; descriptive) selected the same layer and token position every time, with directions at cosine 0.98 to 0.99 to the original (Table 2). Their gains on test nevertheless range from +12.4 to +21.6 points, and the original direction (+20.4) lies near the top of that range. Two of the five gains are below that of NF4 alone (+14.0), so the share of the gain reached by NF4 ranges from 0.65 to 1.13 across the six directions. Every direction still exceeds the largest placebo (+5.6) by its point estimate, passes C^{NF4} (0.71 to 1.00) and fails C' (2.41 to 3.46; see below). The registered verdict for the original direction stands, but its margin over NF4 does not survive the sampling of extraction prompts, whereas its margin over random directions does. Taken together, the direction does more than a random perturbation of the same size, but a gain of similar size is available from quantization, which has nothing to do with refusal.

A stricter safety threshold (post hoc). NF4 quantization itself lowers the safety of Qwen2.5-1.5B: it raises the harmful-content rate by 4.6 points on development data and 8.3 on test, and lowers the refusal rate by 6.6 and 14.3 points (descriptive). Anchored to NF4, the safety thresholds therefore admit large harms. With each safety threshold set to the measurement resolution alone (C'; capability thresholds unchanged), `wang:0.8:0.5` fails on both splits, $C' = 2.50$ and 3.31. The deciding vital is the harmful-content rate, which the edit raises by 3.4 and 8.1 points against resolutions of 1.4 and 2.5, and whose judge reached $\kappa = 1.000$ on Qwen2.5-1.5B. Every placebo fails as well (C' from 3.5 to 18), and so does every re-extracted direction. The only unit that passes on Qwen2.5-1.5B is the system prompt ($C' = 0.07$ and 0.17). The Q8_0 model, used as a negative control, passes on every model and split under both indices (C^{NF4} from 0.05 to 0.30, C'

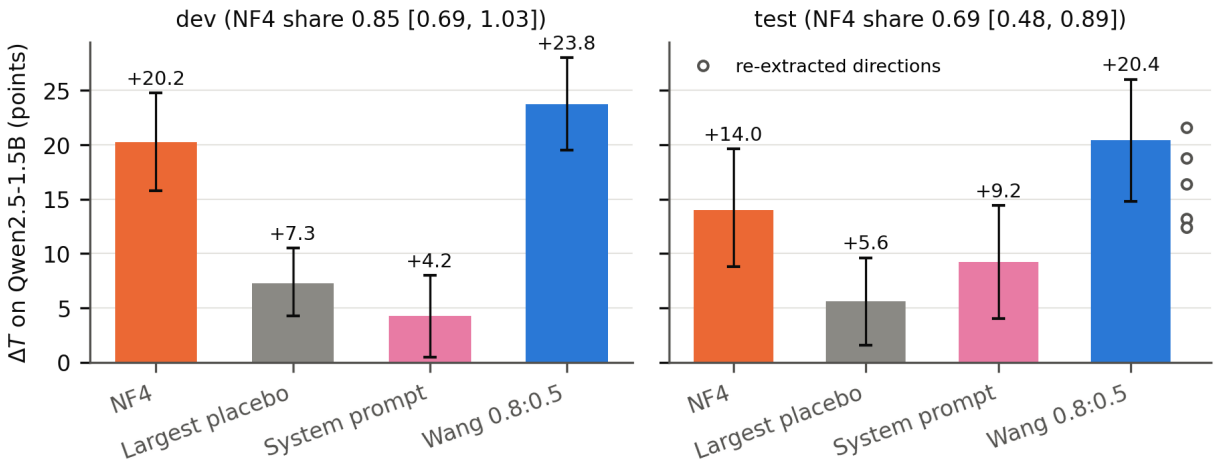


Figure 4: Gain in helpful compliance on Qwen2.5-1.5B from NF4 quantization, the KL-matched placebo with the largest gain, the system prompt and `wang:0.8:0.5` (95% CIs). Titles give the share of the edit's gain reached by NF4 alone (post hoc). Circles in the test panel are the five re-extracted directions.

at most 0.35), so the index does not fail a near-harmless perturbation through noise alone. This control is weak, because Q8_0 changes few answers; it does not bound the rate at which the index would fail a harmless edit that changes many.

6.4. Folding on post-norm models

The γ-compensated fold. On Gemma-2-2B the update produced non-finite weights (Section 3); these units were invalid and were not measured on test. On OLMo-2-1B it collapsed the model: $\Delta T = -54.0$ and -52.0 points on development data and -88.0 and -78.8 on test, with C^{NF4} from 38 to 216.

The pseudo-inverse fold. The rule selected $\varepsilon = 0.01$ on Gemma-2-2B, where the mean deviation of the write from the exact ablation is 0.089% against 0.21% at $\varepsilon = 0$, and $\varepsilon = 0.1$ on OLMo-2-1B (the edge of the grid), with 0.17% against 17.5% (Fig. 5). The fold reproduces the exact ablation: its ΔT differs from that of residual-stream hooks by at most 2.0 points on development data and 1.2 on test (Table 3), and its damage index is close as well (Gemma-2-2B, development: 12.6 against 11.9 at $c = 0.5$; 105 against 100 at $c = 1$). Both are damaged configurations, so this is a check that the fold implements the ablation, not evidence that the ablation is useful.

The failures come from the ablation, not from folding. Every exact ablation (hooks on the residual-stream writes, or the pseudo-inverse fold) has C^{NF4} between 4.4 and 145 on Gemma-2-2B and OLMo-2-1B. On OLMo-2-1B at $c = 1$ the exact ablation destroys capabilities (MMLU drops by 10.8 points on development data; $\Delta T = -48.2$ and -55.6), while the naive fold, which leaves the direction in the residual stream, scores 54.5 and 50.4 points higher. The choice of fold changes no verdict: no variant is acceptable on either model.

Table 3: Mechanism comparisons at $\lambda = 1$: paired differences in ΔT (points, 95% CI). Every compared configuration has $C^{\mathrm{NF4}} > 1$; the pseudo-inverse fold is compared with the residual-stream ablation as a check of implementation fidelity, for which C is also reported in the text.

Model	Pair (c)	dev	test
Gemma-2-2B	naive fold − residual hooks (1)	+24.5 [+19.5, +29.5]	−0.8 [−4.4, +2.8]
	naive fold − residual hooks (0.5)	−7.2 [−10.5, −4.0]	−4.4 [−7.6, −1.6]
	released hooks − residual hooks (1)	+5.5 [+1.5, +9.5]	+1.2 [−1.2, +3.6]
	released hooks − residual hooks (0.5)	–	+0.4 [−2.8, +3.6]
	naive fold − released hooks (1)	+19.0 [+13.8, +24.2]	−2.0 [−5.2, +1.2]
	naive fold − released hooks (0.5)	–	−4.8 [−8.4, −1.6]
	pseudo-inverse fold − residual hooks (1)	+1.0 [−2.8, +4.8]	+0.8 [−1.6, +3.2]
	pseudo-inverse fold − residual hooks (0.5)	−0.8 [−2.0, +0.5]	−1.2 [−3.2, +0.4]
	pseudo-inverse fold − naive fold (1)	−23.5 [−28.5, −18.5]	+1.6 [−2.0, +5.2]
	pseudo-inverse fold − naive fold (0.5)	+6.5 [+3.2, +9.8]	+3.2 [+0.8, +5.6]
OLMo-2-1B	naive fold − residual hooks (1)	+54.5 [+49.5, +59.5]	+50.4 [+44.0, +56.4]
	naive fold − residual hooks (0.5)	−1.2 [−3.8, +1.2]	−0.4 [−2.0, +0.8]
	released hooks − residual hooks (1)	+35.2 [+30.2, +40.5]	+34.8 [+27.6, +41.6]
	released hooks − residual hooks (0.5)	−6.2 [−10.5, −2.0]	−8.0 [−12.0, −4.0]
	naive fold − released hooks (1)	+19.2 [+14.0, +24.5]	+15.6 [+10.4, +20.8]
	naive fold − released hooks (0.5)	+5.0 [+0.5, +9.5]	+7.6 [+3.6, +11.6]
	pseudo-inverse fold − residual hooks (1)	+2.0 [−0.5, +4.5]	+0.0 [−5.6, +5.6]
	pseudo-inverse fold − residual hooks (0.5)	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
	pseudo-inverse fold − naive fold (1)	−52.5 [−57.5, −47.5]	−50.4 [−56.8, −43.6]
	pseudo-inverse fold − naive fold (0.5)	+1.2 [−1.2, +3.8]	+0.4 [−0.8, +2.0]

Fold against hooks. On development data the naive fold exceeds the released hooks at $c = 1$ by 19.0 points on Gemma-2-2B and 19.2 on OLMo-2-1B. On test the difference replicates on OLMo-2-1B (+15.6) but not on Gemma-2-2B (−2.0 [−5.2, 1.2]), where XSTest leaves little headroom. On Gemma-2-2B this difference also includes the effect of the tied output head; the pseudo-inverse and naive folds, which share it, differ by −23.5 points [−28.5, −18.5] at $c = 1$ on development data and by +1.6 [−2.0, 5.2] on test. On the pre-norm Qwen2.5-1.5B, where the released hooks implement the ablation, the fold still exceeds them by 9.8 points (development) and 5.2 (test); the tied output head contributes 3.5 points [1.8, 5.5] of the difference at `wang:0.8:0.5` (development only), measured by folding the output matrices alone and ablating the embedding with a hook. Because every compared configuration on the post-norm models has $C^{\mathrm{NF4}} > 1$, these differences describe the mechanism, not usable edits.

6.5. Truthfulness

Reproduction. Scored with the procedure of their released code and the registered tokenization, which starts each sequence with one BOS token, our reimplementation gives the unedited Gemma-2-2B 212 correct test answers of 409 (51.8%) and the published configuration 237 (57.9%). The published values are 233 of 409 for that configuration (read from their table) and about 52.8% for the unedited model (read from their figure). Their released code adds the BOS token a second time. Scored on sequences that start with two BOS tokens, a variant our code computed alongside the registered one before any result but that the pre-

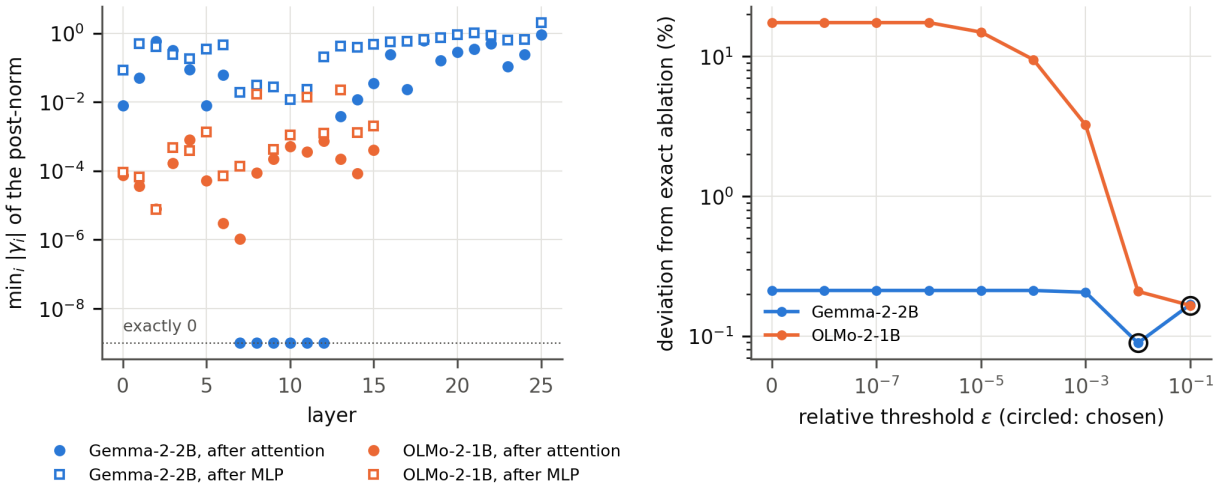


Figure 5: Left: smallest gain $|\gamma_i|$ of each post-sublayer norm (Gemma-2-2B: $1 + w$; OLMo-2-1B: w); points on the dotted line are exactly zero. Right: mean relative deviation of the pseudo-inverse fold's write from the exactly ablated write, against the threshold ε (0 at the left); circles mark the values chosen by the pre-registered rule.

registration does not name, the counts are 216 (52.8%) and 233 (57.0%), matching both published values. This suggests, without establishing it, that the published scores were computed with a repeated BOS token; we did not compare answers item by item. Utility reproduced within the pre-registered tolerance overall (41.7 against 44.4 ± 3), but not on GSM8K (21.6 against 27.2 ± 5; Appendix A, S5).

Metric floor. With the scoring used in the original evaluation, corrupted models can score above intact ones. On the training split of a smaller model, Qwen2.5-0.5B, a model with zeroed attention outputs reaches 53.9% length-balanced accuracy against 45.6% for the intact model; on the TruthfulQA test split (descriptive), a fully re-initialized Gemma-2-2B scores 55.3% against 51.8%, a difference of about one standard error. With length-balanced PMI, every corrupted model on the training split stays between 42.9% and 50.3%, at or below chance, and this metric was adopted. No metric-floor check of the original scoring was run on the Gemma-2-2B training or development data.

Development. On the 158 development questions, none of the 23 configurations has a ΔT whose interval excludes zero; the point estimates range from -3.9 to $+3.9$ points, with a standard error of about 3.6. The published configuration gains 3.7 points [-3.6, 10.8] and lowers MMLU by 7.0 points [3.8, 10.2] and GSM8K accuracy by 30 points. On the already inspected test split, the published configuration gains 4.8 points [0.6, 9.1] (descriptive). Detecting a gain of 4 points would need roughly 400 to 1,000 questions; we therefore report no detectable gain at preserved capabilities, not an absence of effect.

6.6. A system prompt as baseline

The fixed system prompt raises T on Qwen2.5-1.5B by 4.2 points [0.5, 8.0] on development data and 9.2 [4.0, 14.4] on test, with $C^{\mathrm{NF4}} = 0.02$ and 0.17; it is the only unit that passes the stricter post-hoc threshold. On Gemma-2-2B it leaves T unchanged (+0.8, −2.8) and fails ($C^{\mathrm{NF4}} \approx 40$, through the end-of-turn vital; 4.3 to 4.5 without it), and on OLMo-2-1B it lowers T (−15.0, −9.2). On the one model where an edit appeared to work, a system prompt gave a smaller gain without measurable damage.

7. Discussion and limitations

Scope of the conclusions about the evaluated methods. Wang et al. (2025) and Sun et al. (2026) evaluated other, mostly larger models, and our results do not test their reported results on those models. For Steer2Edit we reproduced the reported truthfulness score on Gemma-2-2B with the original scoring; what changes is the reading of that score once the metric is checked against corrupted models and capabilities are held fixed. For the over-refusal method, our results concern three small models, two of which normalize sublayer outputs, a setting the method was not designed or evaluated for.

Quantization anchors suit capabilities, not safety. Normalizing harm by the harm of weight quantization gives a threshold tied to a perturbation practitioners accept, and on capabilities it behaved as intended: the Q8_0 model passed on every model and split. On safety it can be lenient. On Qwen2.5-1.5B, NF4 quantization itself raised the harmful-content rate by 8.3 points on test, so an NF4-anchored threshold admitted an edit with a harm of the same size; with thresholds at the measurement resolution the edit failed. This differs from the finding that 4-bit GPTQ and AWQ quantization largely preserves the trustworthiness of 13B models (Hong et al., 2024); our model is much smaller, our format is NF4, and we did not investigate the cause. We recommend anchoring capability vitals to quantization and setting safety thresholds by measurement resolution alone, or anchoring them to an 8-bit format.

Controls for specificity. A behavioral gain that quantization reproduces for the most part is weak evidence of a targeted edit, even when it exceeds random directions of the same size. Quantization and KL-matched placebos cost two extra evaluations and should accompany any claim that a weight edit acts on a behavior. The comparison should be repeated over several extractions of the direction: on Qwen2.5-1.5B the sample of extraction prompts alone moved the gain by 9.2 points and reversed its order relative to quantization.

Sample size and metric floors. The safety verdict on Gemma-2-2B turned on the number of harmful prompts: with 200, harms below about 2.2 points cannot be resolved. A safety set should be sized so that its resolution is below the threshold it is meant to test. Similarly, a target metric should be checked against corrupted models before it is used: in both of our cases a natural metric could reward broken models.

Folding. On post-norm models a naive fold is a different edit from the ablation it is meant to implement, and the γ-compensated fold can fail on real checkpoints whose normalization gains contain zeros or near-zeros. Proposition 2 gives a condition, checkable from r and Γ alone, under which folding is exact, and the pseudo-inverse fold implements the ablation closely when it is not. Folding did not change any verdict here, because the ablation itself failed on both post-norm models.

Limitations. The specificity results rest on one model, on which C is descriptive because the refusal judge failed validation, and the re-extraction was run for one configuration. The models are small (1 to 2B parameters) and were run in fp32 on one type of GPU; NF4 is not the 4-bit format most local deployments use (for example Q4_K_M). Reference labels come from two LLM annotators adjudicated by the author, with a 30-item human check. Every configuration was run once with one seed. XSTest leaves little headroom on Gemma-2-2B and OLMo-2-1B, and the Gemma-2-2B fold–hook difference seen on development data did not replicate on test. The truthfulness case has low power, its test split had been seen, and its metric-floor check of the original scoring used a smaller model. The perplexity criterion of the degeneration gate was applied to benign replies only. ProSafePrune (Chen et al., 2026) could not be compared because no code was released. The stricter safety threshold, the share of the gain reached by NF4, the Q8_0 negative control and the sample-size analysis are post hoc.

8. Conclusion

A pre-registered protocol with metric floors, quantization-anchored damage, validated judges and adequately sized safety sets changes what can be concluded about behavioral weight edits. The over-refusal method we examined has no acceptable configuration on two of three small models. On the third, its direction does more than a random perturbation of the same size, but quantization alone gives a gain of similar size, and every extraction of the direction fails a stricter safety threshold that a system prompt passes. For truthfulness, the scoring of prior work can reward broken models, and no gain is detected at preserved capabilities. On models with post-sublayer normalization, folding an ablation requires care, and a pseudo-inverse fold implements it closely. Code, configurations and results are released so that the protocol can be applied to other edits.

Appendix A. Deviations from the pre-registration

Every deviation is recorded with its time in the pre-registration file. Times are local (UTC+7), 2026.

S1 Reference labels were produced by two LLM annotators and adjudicated by the author, instead of by the author alone (31 Oct, 13:40, after the LLM labels and before any judge ran); a blind 30-item human check was added.

S2 The harmful development set was enlarged from 200 to 1,031 prompts and the refusal score was added as a vital (31 Oct, 15:24), after the feasibility result on Gemma-2-2B was seen to depend on a single reply, and before any new measurement. This made the conclusion stricter.

S3 The perplexity criterion of the degeneration gate was dropped for harmful replies to save compute (31 Oct, 15:50, before any result of that job).

S4 OR-Bench-Hard was split at random, stratified by category, rather than by topic as in Wang et al. (2025) (30 Oct, before running); XSTest tests transfer.

S5 The Steer2Edit reproduction missed its GSM8K tolerance (21.6 against 27.2 ± 5); the reproduction was accepted with this note (30 Oct).

S6 The scope was changed to the present study (31 Oct, 18:17); an agent and a factorial test planned earlier were moved to later work.

S7 The TruthfulQA test split was seen during the reproduction; its numbers are descriptive (30 Oct).

S8 The γ-compensated fold was not measured on the Gemma-2-2B test split, because it yields non-finite weights (1 Nov, 16:38, before the test run).

S9 The rule required re-selecting a judge that failed on a new model. For the refusal judge on Qwen2.5-1.5B we instead reported every dependent quantity as descriptive (1 Nov, about 19:00, after the agreement was seen and before any test result).

S10 ε for Gemma-2-2B was computed on GPU inside the job rather than on CPU before it, with the same code, grid and criterion (1 Nov, 18:55, before the job).

S11 Four pseudo-inverse units were judged in a separate job after a crash of the judging step, with the same judge on the saved replies (2 Nov, 00:10).

Post-hoc analyses, proposed after the corresponding results were seen: the safety-set size analysis (Fig. 3); the share of the gain reached by NF4 (Fig. 4); the stricter safety threshold C'; the Q8_0 negative control; C without the end-of-turn vital. The re-extraction of directions on Qwen2.5-1.5B (Table 2) was proposed in review and registered before it was run (2 Nov, 22:00); its results are descriptive. All tables and figures were recomputed by one script with new bootstrap seeds (Section 4).

CRediT authorship contribution statement

Manh Pham Huu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Table A.4: Over-refusal, development split (as Table 1).

Edit	ΔT [95% CI]	C^{NF4}	C^{Q8}	C' (post hoc)	Acc.
Gemma-2-2B (T_0 = 43.2%)					
System prompt	+0.8 [−3.5, +5.0]	39.0	518	39.0	no
Naive fold (λ=1.0, c=1.0)	+29.5 [+24.5, +34.5]	33.0	548	33.0	no
Naive fold (λ=1.0, c=0.5)	+11.5 [+8.2, +15.0]	6.08	101	6.08	no
Naive fold (λ=0.8, c=0.5)	+22.2 [+18.2, +26.5]	17.8	296	17.8	no
Released hooks (λ=1.0, c=1.0)	+10.5 [+4.8, +16.2]	101	1683	101	no
Residual hooks (λ=1.0, c=1.0)	+5.0 [−0.5, +10.8]	100	1663	100	no
Residual hooks (λ=1.0, c=0.5)	+18.8 [+14.5, +23.0]	11.9	197	11.9	no
Pseudo-inverse fold (λ=1.0, c=1.0)	+6.0 [+0.5, +11.5]	105	1748	105	no
Pseudo-inverse fold (λ=1.0, c=0.5)	+18.0 [+13.8, +22.2]	12.6	209	12.6	no
γ-comp. fold (λ=1.0, c=1.0)	−43.2 [−48.0, −38.5]	96.3	278	96.3	no
γ-comp. fold (λ=1.0, c=0.5)	−43.2 [−48.0, −38.2]	96.3	278	96.3	no
Qwen2.5-1.5B (T_0 = 25.0%)					
NF4 (anchor)	+20.2 [+15.8, +24.8]	1.00	465	20.3	–
Q8_0 (anchor)	−0.5 [−2.0, +0.8]	0.05	0.61	0.05	–
System prompt	+4.2 [+0.5, +8.0]	0.02	0.50	0.07	yes†
Naive fold (λ=1.0, c=1.0)	+16.5 [+11.5, +21.5]	2.01	936	40.8	no
Naive fold (λ=1.0, c=0.5)	+4.5 [+1.5, +7.5]	0.63	293	12.8	yes†
Naive fold (λ=0.8, c=0.5)	+23.8 [+19.5, +28.2]	0.74	19.5	2.50	yes†
Released hooks (λ=1.0, c=1.0)	+6.8 [+2.5, +11.0]	1.39	647	28.2	no
Released hooks (λ=1.0, c=0.5)	−0.8 [−4.5, +3.2]	1.03	299	13.0	no
Placebo 4	+7.2 [+4.2, +10.5]	0.91	421	18.4	yes†
OLMo-2-1B (T_0 = 54.0%)					
NF4 (anchor)	+0.2 [−3.0, +3.5]	1.00	56.6	1.40	–
Q8_0 (anchor)	−0.2 [−1.8, +1.2]	0.19	1.00	0.19	–
System prompt	−15.0 [−19.5, −10.5]	1.01	7.00	1.01	no
Naive fold (λ=1.0, c=1.0)	+6.2 [+1.5, +11.0]	33.0	257	33.0	no
Naive fold (λ=1.0, c=0.5)	+4.8 [+2.0, +7.8]	13.4	69.6	13.4	no
Naive fold (λ=0.8, c=0.5)	+11.0 [+7.5, +14.8]	18.2	94.1	18.2	no
Released hooks (λ=1.0, c=1.0)	−13.0 [−19.0, −7.0]	46.7	396	46.7	no
Released hooks (λ=1.0, c=0.5)	−0.2 [−5.0, +4.5]	41.8	285	41.8	no
Residual hooks (λ=1.0, c=1.0)	−48.2 [−53.2, −43.2]	145	985	145	no
Residual hooks (λ=1.0, c=0.5)	+6.0 [+2.8, +9.2]	18.9	98.0	18.9	no
Pseudo-inverse fold (λ=1.0, c=1.0)	−46.2 [−51.2, −41.2]	126	933	126	no
Pseudo-inverse fold (λ=1.0, c=0.5)	+6.0 [+2.8, +9.5]	18.9	97.6	18.9	no
γ-comp. fold (λ=1.0, c=1.0)	−54.0 [−58.8, −49.0]	216	1712	216	no
γ-comp. fold (λ=1.0, c=0.5)	−52.0 [−57.0, −46.8]	160	830	160	no

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code, configurations (row indices and content hashes of every data split), results, and reference labels are available at Zenodo, `https://doi.org/10.5281/zenodo.23909304`, under the Apache-2.0 licence (code) and CC BY 4.0 (data). Model replies to harmful prompts are not released. Third-party prompts are not redistributed; scripts rebuild them from their sources.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to draft and edit the text, to write and run the analysis and experiment code (Claude Code), and to critique the study design and the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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